# Algebra II with Trigonometry Chapter 6.3

**Olympiad Solutions**

Saved tutoring response with MathJax rendering. Original PDF file: 4531393_alg-2trig-h-chp-6-3-note.pdf

## Solutions

---

**1. Let $f(x) = \sin x + \sin(2x) + \sin(3x)$. What is the maximum value of $f(x)$?**

Let’s use sum-to-product identities. First, combine $\sin x + \sin(3x)$:

$$\sin x + \sin(3x) = 2\sin(2x)\cos x$$

So,

$$f(x) = 2\sin(2x)\cos x + \sin(2x) = \sin(2x)(2\cos x + 1)$$

Let’s write everything in terms of $\sin x$:

But a better strategy: set $u = \sin x$, the maximum of $\sin(2x)$ is 1 and minimum is -1, so consider when $2\cos x + 1$ is positive.

But to maximize $f(x)$, set the derivative to zero, but let's look for a more elegant way.

Let’s use:

$$f(x) = \sin x + \sin(2x) + \sin(3x) = [\sin x + \sin(3x)] + \sin(2x) = 2\sin(2x)\,\mathrm{co}$$

Let’s denote $y = \sin(2x)(2\cos x + 1)$. Let's try $\sin(2x) = 1$:

If $\sin(2x) = 1 \Rightarrow 2x = \frac{\pi}{2} \Rightarrow x = \frac{\pi}{4}$:

Then, $\cos x = \cos\frac{\pi}{4} = \frac{\sqrt{2}}{2}$:

$$f\left(\frac{\pi}{4}\right) = 1 \cdot \left(2 \cdot \frac{\sqrt{2}}{2} + 1\right) = 1 \cdot (\sqrt{2} + 1) = \sqrt{2} + 1 \approx 2.414$$

Try $\sin(2x) = -1$:

If $x = \frac{3\pi}{4}$:

$\cos x = \cos\frac{3\pi}{4} = -\frac{\sqrt{2}}{2}$

$$f\left(\frac{3\pi}{4}\right) = -1 \cdot (2 \cdot (-\frac{\sqrt{2}}{2}) + 1) = -1 \cdot (-\sqrt{2} + 1) = \sqrt{2} - 1 \approx 0.414$$

Try $2\cos x + 1 = 0 \Rightarrow \cos x = -\frac{1}{2}$:

At $x = \frac{2\pi}{3}$, $\sin(2x) = \sin(\frac{4\pi}{3}) = -\frac{\sqrt{3}}{2}$

So, $f(x) = -\frac{\sqrt{3}}{2} \cdot 0 = 0$.

Thus, the maximum is at $x = \frac{\pi}{4}$: $\boxed{1 + \sqrt{2}}$ is the maximum value.

---

**2. Suppose $y = \cos(kx)$ has period $\frac{\pi}{3}$. Without solving, what is the value of $k$?**

The period of $\cos(kx)$ is $T = \frac{2\pi}{k}$. Set this equal to $\frac{\pi}{3}$:

$$\frac{2\pi}{k} = \frac{\pi}{3} \implies k = 6$$

$\boxed{6}$

---

**3. For which value(s) of $c$, is the function $y = \sin(x) + c\sin(x + \pi/3)$ an even function?**

Recall: A function is even if $f(-x) = f(x)$.

$$f(x) = \sin x + c\sin(x + \pi/3)$$

$$f(-x) = \sin(-x) + c\sin(-x + \pi/3) = -\sin x + c\sin(\pi/3 - x)$$

So, require $f(-x) = f(x)$:

$$-\sin x + c\sin(\pi/3 - x) = \sin x + c\sin(x + \pi/3)$$

But $\sin(\pi/3 - x) = \sin(x + \pi/3)$ only if $x = -x$, but generally not for all $x$
.

Let's use the sum formula:

$$\sin(\pi/3 - x) = \sin(\pi/3)\cos x - \cos(\pi/3)\sin x = \frac{\sqrt{3}}{2}\cos x - \frac{1}{2}\sin x$$

$$\sin(x + \pi/3) = \sin x\cos(\pi/3) + \cos x\sin(\pi/3) = \frac{1}{2}\sin x + \frac{\sqrt{3}}{2}\cos x$$

Plug in:

$$-\sin x + c\left(\frac{\sqrt{3}}{2}\cos x - \frac{1}{2}\sin x\right) = \sin x + c\left(\frac{1}{2}\sin x + \frac{\sqrt{3}}{2}\cos x\right)$$

Collect $\sin x$ and $\cos x$ terms:

LHS:

$$\left[-\sin x - \frac{c}{2}\sin x\right] + \left[c\frac{\sqrt{3}}{2}\cos x\right]$$

RHS:

$$\left[\sin x + c\frac{1}{2}\sin x\right] + \left[c\frac{\sqrt{3}}{2}\cos x\right]$$

Rearrange: Coefficients of $\sin x$ must match:

LHS: $-\sin x - \frac{c}{2}\sin x$, RHS: $\sin x + \frac{c}{2}\sin x$

Set equal:

$$-1-\frac{c}{2}=1+\frac{c}{2} \implies -1-\frac{c}{2}=1+\frac{c}{2} \implies -2=c$$

Thus, $\boxed{c=-2}$.

---

**4. Let $A=\sin^2\left(\frac{\pi}{8}\right)+\sin^2\left(\frac{3\pi}{8}\right)+\sin^2\left(\frac{5\pi}{8}\right)+\sin^2\left(\frac{7\pi}{8}\right)$.**

**Compute $A$ mentally.**

Note the symmetry:

$\sin^2(\theta)+\sin^2(\pi-\theta)=\sin^2\theta+\sin^2\theta=2\sin^2\theta$, since $\sin(\pi-\theta)=\sin\theta$

But check:

$$\sin^2\left(\frac{\pi}{8}\right)+\sin^2\left(\frac{7\pi}{8}\right)$$

But $\sin(\frac{7\pi}{8})=\sin(\pi-\frac{\pi}{8})=\sin(\frac{\pi}{8})$ So, sum is $2\sin^2(\frac{\pi}{8})$

Similarly,

$$\sin^2\left(\frac{3\pi}{8}\right)+\sin^2\left(\frac{5\pi}{8}\right)=2\sin^2\left(\frac{3\pi}{8}\right)$$

Because $\sin(\frac{5\pi}{8})=\sin(\pi-\frac{3\pi}{8})=\sin(\frac{3\pi}{8})$

Thus,

$$A=2\sin^2\left(\frac{\pi}{8}\right)+2\sin^2\left(\frac{3\pi}{8}\right)$$

Recall $\sin^2\alpha=\frac{1-\cos(2\alpha)}{2}$:

So,

$$A=2\times\frac{1-\cos(\frac{\pi}{4})}{2}+2\times\frac{1-\cos(\frac{3\pi}{4})}{2}=(1-\cos(\frac{\pi}{4}))+(1-\cos(\frac{3\pi}{4}))$$

$\cos(\frac{\pi}{4}) = \frac{\sqrt{2}}{2}$ $\cos(\frac{3\pi}{4}) = -\frac{\sqrt{2}}{2}$

So,

$$A = [1 - \frac{\sqrt{2}}{2}] + [1 + \frac{\sqrt{2}}{2}] = 1 - \frac{\sqrt{2}}{2} + 1 + \frac{\sqrt{2}}{2} = 2$$

$\boxed{2}$

---

**5. Consider $y = 2\sin x + \sin^2 x$. What is the range of $y$?**

Let $s = \sin x$, with $s \in [-1, 1]$:

$$y = 2s + s^2$$

This is a quadratic in $s$: $y = s^2 + 2s$.

The quadratic opens upward (since coefficient of $s^2$ is positive).

The minimum is at one endpoint OR at $s = -1$, since the vertex is at $s = -1$ (since $s_{min} = -\frac{b}{2a} = -\frac{2}{2} = -1$), and $s \in [-1, 1]$.

So check endpoints:

At $s = -1$:

$$y(-1) = (-1)^2 + 2(-1) = 1 - 2 = -1$$

At $s = 1$:

$$y(1) = (1)^2 + 2(1) = 1 + 2 = 3$$

So range of $y$ is $[-1, 3]$. $\boxed{[-1,\ 3]}$

---

**6. The graphs of $y = a\sin(kx - b)$ and $y = a\cos(kx - b)$ are shifted so that their maximum points coincide for $x > 0$. What must be the distance (in terms of $a, k$) between their first maximums?**

For $y = a\sin(kx-b)$, maxima occur when $\sin(kx-b)=1$:

$$kx - b = \frac{\pi}{2} + 2\pi n \implies x_1 = \frac{\pi}{2k} + \frac{2\pi n}{k} + \frac{b}{k}$$

First max for $x > 0$: set $n = 0$:

$$x_s = \frac{\pi}{2k} + \frac{b}{k}$$

For $y = a\cos(kx-b)$, maxima occur when
$\cos(kx-b) = 1 \implies kx - b = 2\pi m$:

$$x_c = \frac{2\pi m}{k} + \frac{b}{k}$$

First max for $x > 0$: $m = 0$:

$$x_c = \frac{b}{k}$$

So, distance between their first maxima:

$$x_s - x_c = \frac{\pi}{2k}$$

So, the required distance is

$$\boxed{\frac{\pi}{2k}}$$

Note: This answer is independent of $a$, as maxima occur at the same $x$ for any amplitude.